

Capitec Microservice Application

As a Kubernetes administrator you are given access to 3 images, which together form a microservice-based application named: Capitec User Service. This is a simple application used to onboard new users, as leads, so that different business units can pitch products to them. The application has the following components:

- **Postgres database:**
 - Image available from DockerHub
 - Image referenced as: postgres
 - Setup to listen on port: 5432
- **Spring Boot Backend Service**
 - Images available from DockerHub
 - ♣ Windows-compatible image: thapeloeseema/capitec-user-service:backend-amd64
 - ♣ Mac-arm compatible image: thapeloeseema/capitec-user-service:backend-arm64
 - Setup to listen on port: 8080
- **Angular Frontend Service**
 - Images available from DockerHub
 - ♣ **Windows-compatible image:** thapeloeseema/capitec-user-service:frontend-amd64
 - ♣ **Mac-ARM compatible image:** thapeloeseema/capitec-user-service:frontend-arm64
 - Setup to listen on port: 4200

You are tasked with deploying this application on a local Kubernetes cluster such that the front-end application is available from outside of the Kubernetes cluster, the front-end can communicate internally with the backend, and the backend can communicate internally with the database. We need to have 3 instances of the front end always running, 4 instances of the backend, and 2 instances of the databases.

Johannesburg:

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Cape Town:

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Submission Details

Create a fork of the repository <https://github.com/Geeks4LearningJHB/Capitec-User-Service-Submissions> , and name your fork by appending your name and surname to the main repo's name as follows:

- Capitec-User-Service-Submissions-Thapelo-Seema

This is where you must push all the code in your solution, once you have tested it on your machine.

Once done, you must create a merge request to the main repository.

The deadline is 10 March 2025.

You can reach out to thapelo.seema@geeks4learning.com for any queries or assistance.

The working solution will have an externally accessible website where you can add users as shown below:

